Combustion and Flame 210 (2019) 360–373

Contents lists available at ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

A detailed chemical mechanism for low to high temperature oxidation of n-butylcyclohexane and its validation

Yebing Mao, Ang Li, Lei Zhu*, Zhiyong Wu, Liang Yu, Sixu Wang, Mohsin Raza, Xingcai Lu*

Key Laboratory for Power Machinery and Engineering of M. O. E., Shanghai Jiao Tong University, Shanghai 200240, PR China

ARTICLE INFO

Article history:
Received 20 June 2019
Revised 4 September 2019
Accepted 4 September 2019

Keywords:
Kinetic mechanism
n-Butylcyclohexane
Low temperature chemistry
Autoignition
Oxidation

ABSTRACT

A good knowledge of the reaction kinetics of n-butylcyclohexane, a widely used surrogate component for jet fuels and diesel, is crucial for understanding the combustion chemistry of practical fuels. This paper proposes a detailed kinetic model consisting of 1802 species and 7246 reactions to describe from low to high temperature chemistry of n-butylcyclohexane. The autoignition and oxidation of this fuel were investigated over a wide range of conditions. Ignition delay times were measured at 10, 15 and 20 bar for equivalence ratios of 1 and 1.5 under highly diluted conditions in a heated rapid compression machine. Oxidation experiments were performed in a flow reactor for equivalence ratios of 1 and 1.5 over the temperature range of 650–1075 K at 1 atm. Several oxidation products were identified and their mole fraction profiles were measured as a function of temperature. The model was evaluated with respect to the new experimental data. Additionally, the model was compared to experimental data in the literature, including ignition delay times in both rapid compression machine and shock tube, speciation profiles in jet-stirred reactor. Good agreement was achieved with all the data. Reaction flux analyses and sensitivity analyses were performed in order to provide insight into the combustion kinetics of n-butylcyclohexane. The present model can be used to construct the kinetic models for surrogates of jet fuels or diesel, as well as to work as the starting mechanism for the development of kinetic models of larger alkylcyclohexane and bicyclic cyclohexane fuels.

1. Introduction

The integration of chemical reaction kinetic mechanisms and computational fluid dynamics provides the promise of optimizing engine design and combustion technology [1]; therefore, there has been an increasing interest in developing accurate kinetic models for transport fuels in recent decades. The promise of reactivity-controlled combustion concepts, further motivates the quest for a good understanding of combustion chemistry of fuels. The modeling in engines typically uses surrogate fuels to overcome the challenge of the complexity of real fuels. This requires reliable kinetic models of neat surrogate components. Cycloalkanes are a very important constituent family in real transport fuels [2–5] and are widely used in the surrogate fuel formulation [6–11]. Cycloalkanes are also believed to play an important role in the formation of soot precursor [12–15] and serve as heat sinks for aeroengines. Therefore, considering the great relevance of these cyclic hydrocarbons to combustion processes, there is a necessity to conduct detailed studies for cycloalkanes.

Although the combustion chemistry of cycloalkanes has been an issue of interest in recent years, the detailed investigations on cycloalkanes, especially the alkylcyclohexanes with long substitute chain, are scarce compared with the extensively studied alkanes. n-Butylcyclohexane (NBCH) is a good compromise between a longer chain (required by jet fuel and diesel surrogates) and reasonable molecular size, and thus is usually selected as a representative component of cycloalkanes in surrogate fuels [16–18]. Inconsistent with its importance, the studies on NBCH are far from sufficient due to its large molecular weight and complex molecular structure. For the time being, to our knowledge, there are only two kinetic models proposed to simulate the oxidation of NBCH, namely JetSurF model [19,20] and the combined model proposed by Natelson et al. [21]. JetSurF2.0 is the latest version of the JetSurF model proposed by Wang et al. [19] in an ongoing effort towards a jet fuel surrogate mechanism. Regretfully, this model is an 'untuned work-in-progress' and includes only limited low-temperature chemistry, and thereby requires further work. Natelson et al. [21] extended the JetSurF 1.1 [20] to describe the low temperature oxidation of NBCH by incorporating a reaction scheme of 42 species and 80 reactions, and validated this combined model against their flow reactor experiments. To our knowledge, this is the first exploration of the low temperature chemistry for NBCH. However, in a very

* Corresponding authors.
E-mail addresses: tonyzhulei@sjtu.edu.cn (L. Zhu), lyuxc@sjtu.edu.cn (X. Lu).

https://doi.org/10.1016/j.combustflame.2019.09.007

Table 1
Summary of experimental studies on NBCH.

Reactor	*T* (K)	*P*	Equivalence ratio	Type of data	Reference
ST	1280–1480	1.5, 3 bar	0.45, 0.83/0.88	IDTs	[23]
ST	707–1458	2, 5, 15 bar	0.5, 1, 2	IDTs	[22]
JSR	500–1100	1.067 bar	0.25,1, 2	Species profile	[24]
Counterflow configuration	353	1 atm	0.7–1.5	Laminar flame speed	[25]
Counterflow configuration	373	1 atm	–	Ignition temperature	[26]
Flow reactor	600–820	8 atm	0.38	Species profiles	[21]
ST and RCM	612–1374	10, 15, 20 bar	0.5, 1, 1.5	IDTs	[27]

recent study by Zhang et al. [22], they compared the prediction from Natelson's model with their measured ignition delay times (IDTs) in a shock tube (ST) and observed a noticeable discrepancy between the predictions and experimental results in both intermediate and high temperature range. Therefore, just as Natelson et al. [21] pointed out, the extensive modification would be needed for the combined model based on new experimental data. Table 1 presents the previous experimental studies on the oxidation of NBCH, a majority of which were obtained after the development of the above-mentioned models, and thus these models were not validated against all the available data yet. Note that the base model (JetSurF model) is only validated against limited experimental data of NBCH, and a semi-lumped low temperature reaction scheme of 80 reactions is difficult to accurately describe the very complex low temperature chemistry. Therefore, simple modification of reaction rate coefficients perhaps is not enough to simulate all the available fundamental combustion experiments, and a more detailed description of the oxidation of NBCH would be needed to improve the performance of the mechanism.

It can be found that the available experimental datasets in Table 1 are still limited, especially the distribution of speciation products of oxidation. To our knowledge, there were only two related studies, one was carried out in a jet-stirred reactor at Nancy [24], and the other in a flow reactor at Drexel University [21] but only covering five temperatures.

In summary, further experimental and modeling efforts are needed in order to improve the understanding of the combustion chemistry of NBCH. Therefore, the objectives of the present study are to (1) experimentally investigate the autoignition and oxidation characteristics of NBCH; and (2) develop a detailed kinetic model to describe the oxidation of NBCH and evaluate its performance against experimental data obtained in this study and available in literature. In the present study, the IDTs of NBCH under highly diluted conditions were measured using a rapid compression machine (RCM) in the low-to-intermediate temperature range, and the oxidation of NBCH was investigated from low to high temperatures in a flow reactor. A detailed kinetic model of 1802 species and 7246 reactions was developed and compared to available experimental data for NBCH in literatures, as well as the present experimental data. Kinetic analyses, including reaction flux analysis and sensitivity analysis, were performed to provide insight into the combustion chemistry of NBCH.

2. Methods

2.1. Ignition delay time measurement

The IDTs of NBCH were measured in the RCM at Shanghai Jiao Tong University (SJTU). The details of the experiments, including the structure of the apparatus, the determination of the temperature at the top dead center and mixture preparation, as well as the experimental uncertainties, had been described in our previous studies [28–35], and will not be repeated here. The IDT in the measurements referred to the time from the end of compression

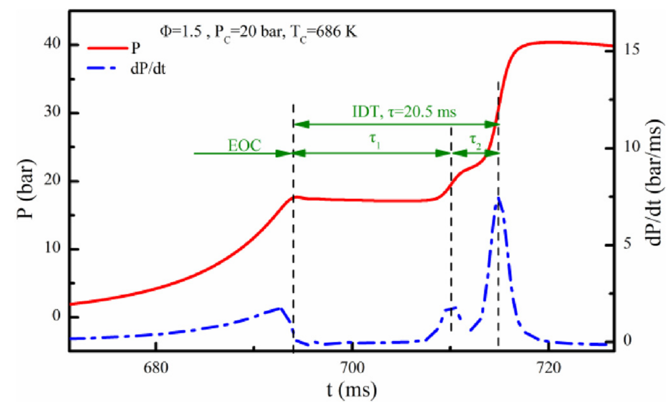


Fig. 1. Typical pressure trace of RCM measurements for NBCH.

Table 2
Test conditions used in the RCM measurements for NBCH.

Equivalent ratio	x_{fuel} (mol%)	x_{O2} (mol%)	x_{N2} (mol%)	Pressure (bar)
1.5	0.695	6.95	92.35	10/15/20
1.0	0.46	6.97	92.57	10/15/20

(EOC) to the maximum dP/dt, as defined in Fig. 1. The systematic uncertainty of the measured IDTs is estimated to be ±10% using the method described in detail in [34].

In the present study, a preheating temperature of 120 °C was used to vaporize the fuel. The test conditions used in the experiments are listed in Table 2.

2.2. Flow reactor oxidation

The flow reactor experiments in this study were performed in the laminar flow reactor at SJTU. The details of this apparatus can be found in previous literature [36,37], and it is briefly introduced here. This system can be divided into three parts, viz, inlet section, reaction section and detection section. The inlet section includes a high-precision syringe pump, a quartz nebulizer, a mixing chamber and mass flow meters to feed the reactants at atmospheric pressure. Before arriving at the reactor, the liquid fuel is preheated to vaporize and mixed with oxygen and inert gas in the mixing chamber to form homogenous mixtures. The reaction section mainly comprises the tubular reactor and the heating furnace. The reactor used in this study is a quartz tube with an inner diameter of 6 mm and effective heating length of 460 mm. It is positioned in the centerline of the cylindrical heating furnace, which allows heating temperature to an upper limit of 1300 K. The online detecting system includes a flame ionization detector (GC-FID, GC2014 SHIMAZU) for detecting hydrocarbons of C1–C5, and a gas chromatography with a thermal conductivity detector (GCTCD, GC2014 SHIMAZU) for detecting H_2, CO_2, CO and O_2.

Oxidation experiments of NBCH were performed at 1 atm for equivalence ratio of 1 and 1.5 over a temperature range of

Table 3
Test conditions of NBCH flow reactor experiments.

Equivalence ratio	Mole fraction (%)		
	χ_{fuel}	χ_{O2}	χ_{N2}
1.0	0.69	10.35	88.96
1.5	0.69	6.90	92.41

650–1075 K. The temperature profiles along the tubular reactor were measured in separate experiments for every set temperature to avoid any wall catalytic reactions between the mixtures and the thermocouple surface. A gas stream of Ar, which was used as the diluent gas and accounted for the majority mole fraction of the mixture in the experiments, was used to replace the mixture in these calibration test. The total flow rate was held constant at 0.0019 mol/s. The specific compositions of the test mixtures are shown in Table 3.

2.3. Simulation methods

The simulations in the present work were performed using the Chemkin software. For the simulation of IDTs, the Closed Homogeneous Batch Reactor model was used along with the proposed kinetic model. For both RCM and ST measurements, we assumed the constant volume constraint (CONV simulation), while for the newly RCM measurements, variable volume simulation, which used the variable volume profiles derived from the non-reactive runs to mimic the heat loss [38], was performed as well. Flow reaction experiment was simulated using the Plug Flow Reactor model, while JSR measurement was simulated using the Perfectly Stirred Reactor model.

3. Chemical kinetic mechanism development

The oxidation of cycloalkanes follows a similar reaction pathway to the alkanes. During the development of the model, the reaction classes proposed by Sarathy et al. [39,40] for alkanes were adopted. The proposed mechanism was developed on the top of a recent published methylcyclohexane (MCH) mechanism [41], of which the core mechanism adopts the AramcoMech model [42,43]. In our NBCH sub-mechanism, the rate constants were taken from the experimental measurements or quantitative calculations in literature or previous kinetic models if available, or estimated based on analogies. The proposed model consists of 1802 species and 7246 reactions, and the full mechanism can be found in Supplemental material. In the following section, the sub-mechanism of NBCH will be introduced in more detail.

3.1. High-temperature chemistry

To the best of our knowledge, JetSurF models [19,20] can be used to describe the high temperature chemistry of NBCH, and has been validated against several experimental datasets at high temperature. Therefore, most reactions of the high temperature reaction scheme were taken from JetSurF models, yet several modifications were made based on some recent works, as introduced below.

The consumption of the parent fuel is initiated from the unimolecular decomposition and H abstraction by oxygen molecules. The unimolecular decomposition of NBCH includes two dissociation channels, viz, C–C bond fission and C–H bond fission. The C–C bond fission of NBCH produces various product radicals via seven different homolytic bond fissions. As a result, among the seven decomposition pathways considered, reaction pathways involving the broken of C–C bonds in the ring lead to the formation of biradical intermediates via ring opening. In this study, the decomposition of the C–C bonds in the side chain was taken from JetSurF 2.0 [19]. As for the scission of cyclic C–C bonds and the subsequent isomerization of the biradicals to alkenes, their reaction rate constants were theoretically calculated at 800–2100 K with an ab initio approach by Ali et al. in a recent study [44]. Hence, these rate constants, expressed in the Plog format, were adopted in the present model. Besides, C–H beta scission channel of the derived biradicals was also included in the present model, and the rate constants were given according to the rate rules summarized by Guo et al. [45]. Typically, the C–H bond fission channels are of little importance compared with C–C decomposition channels due to their high activation energy and the low entropy of the variational transition states [13]. However, these channels are still included in this model. Here, the rate constants in the format of TROE parameters from JetSurF 2.0 were used.

H-atom abstraction reactions are perhaps the most important consumption channel of NBCH. The bimolecular initiation reactions, namely H abstraction by oxygen molecules, were taken from JetSurF 2.0. Besides, five other abstractors are also considered, namely, O, OH, H, HO_2 and CH_3. The present model adopted the rate constants from JetSurF 2.0, but the preexponent factors of H-abstraction by HO_2 were increased by a factor of 15 while those by OH were divided by a factor of 4 based on sensitivity analysis to improve performance of the proposed model. The tuning was based on sensitivity analyses to improve the performance of the model in predicting the IDTs, especially in the intermediate temperature. These reactions were chosen to be tuned according to two principles, namely high sensitivity and great uncertainty. Figure 2 shows a comparison of the predictions from the tuned and untuned model, and it is found that the improvement is evident after tuning.

Six $C_{10}H_{20}$ alkenes were formed through the isomerization of the biradicals. In the proposed model, three types of reactions were mainly considered for the consumption of these alkenes, that is, H-atom abstraction, retro-ene decomposition and decomposition via unimolecular scission of the C–C bond adjacent to the double bond site. Since these alkene species were included in the JetSurF 1.1 model, the unimolecular decomposition and retro-ene decomposition reactions were taken from JetSurF 1.1 [20]. For H-atom abstraction of the $C_{10}H_{20}$ alkenes, the same six abstractors for the parent molecule were considered and the rate constants were estimated according to the rate rules proposed by Guo et al. [45].

The reactions of the $C_{10}H_{19}$ cyclic radicals, $C_{10}H_{19}$ alkenyl radicals and the subsequent intermediates were taken from JetSurF model [19], except the C–H bond scission and oxidation of $C_{10}H_{19}$ cyclic radicals, for which the rate constants were assigned according to the rate rule of Guo et al. [45]. For other smaller species produced from the decomposition of fuel and fuel derived radicals, e.g., cyclohexylpropyl and cyclohexylethyl, cyclohexylmethyl etc., the related reactions were taken from multiple source [19,20,46–49].

3.2. Low-temperature chemistry

To date, there is no detailed low temperature mechanism for NBCH. Therefore, the model development in this work mainly focused on the low temperature chemistry. The low temperature reaction scheme for NBCH in the present model was developed according to the reaction pathway suggested by Sarathy et al. [39,40] and Battin-Leclerc and co-workers [24,50] as shown in Fig. 3.

3.2.1. O_2 addition to fuel derived radicals

NBCH includes one alkyl primary carbon, three alkyl secondary carbons, one tertiary carbon, five cyclic secondary carbons. There are slight differences in bond strength and length due to their

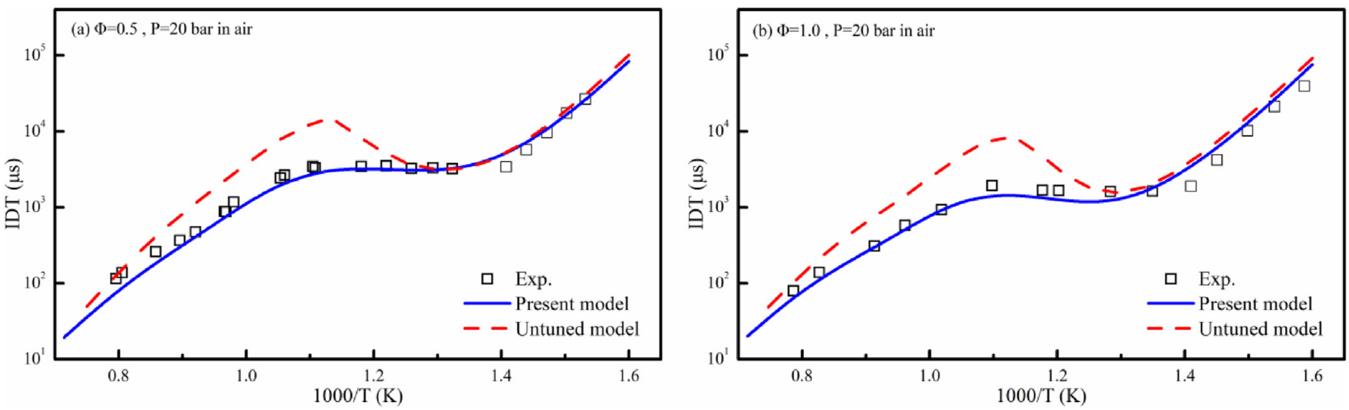


Fig. 2. Evaluation of the untuned and tuned model by comparing their prediction with experimental data in [27].

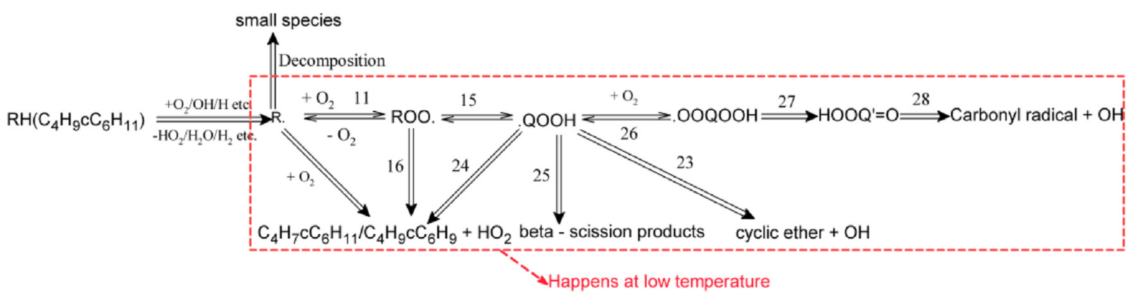
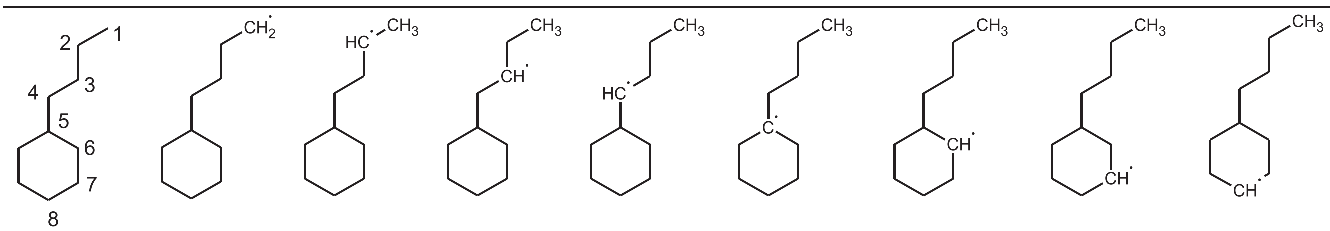


Fig. 3. Low temperature oxidation reaction scheme of NBCH. The numbers on the arrows denote the reaction class.

Table 4
Carbon position for NBCH and nomenclatures of fuel-derived radicals.

C4H9cC6H11	R1	R2	R3	R4	R5	R6	R7	R8



different positions in the molecule. For narrative convenience, they are numbered 1–8 in Table 4. The corresponding eight $C_{10}H_{19}$ cyclic radicals (refer to Table 4 for nomenclature) obtained following the H abstraction from NBCH are also shown in Table 4.

The O_2 addition to fuel derived radical to form alkylperoxy radical (ROO) is the first step of the low temperature chain branching process, and other chain branching reaction classes happen on the premise of its occurrence. Hence, the rate constants of this pathway must be prescribed accurately as much as possible. Some published studies have shown that there are some differences between the low temperature reactivity of cycloalkanes and of alkanes, partly due to the ring strain energy and frozen rotors imposed by the cyclic structure [51–54]. On the other hand, the ring size strongly affects the reactivity of cycloalkanes [55,56]. Therefore, neither the use of alkane analogies nor cycloalkane analogies of different ring size (e.g. cyclopentane) for estimating the reactivity of NBCH are appropriate. Since there were no studies on the addition of oxygen to fuel derived radicals of NBCH, we tried to adopt the rate constants of analogous reactions of cycloalkane with the same ring size (namely, alkyl-cyclohexane). In the present model, the rate constants for the O_2 addition to the cyclic carbon and the primary carbon were estimated based on the analogies of MCH [41], while for the O_2 addition to R2, R3 and R4 (side chain carbons), the rate constants were taken from the analogies from the isomers of pentane according to the carbon type [57].

3.2.2. Isomerization of alkylperoxy radicals to hydroperoxyalkyl radicals

The isomerization of ROO (class 15) is typically considered as one of the most important reaction classes in low temperature, because the subsequent reactions of the produced hydroperoxyalkyl radical (QOOH) are strongly related to the degenerate branching reaction sequence responsible for the reactivity at low temperature ranges.

As a priori model, only one isomerization channel was included for each ROO isomer in Natelson's model. In fact, the reaction scheme of ROO isomerization to QOOH is extremely complicated for NBCH because the combination of the long side chain and ring structure makes many isomerization channels possible. In the present model, isomerizations proceeding via 5, 6 and 7 membered-ring transition states were included. Considering the importance of this reaction class, the accuracy of the rate coefficients would greatly affect the performance of the low temperature reaction scheme, and thereby the rate constant should be prescribed at a high precision level. In Natelson's model, they adopted the rate constants of n-decane from the Milano model [58]. However, Pitz et al. [59] found that it is unreliable to preset the rate constants of cycloalkanes based on knowledge of analogous alkanes. According to Curran et al. [60], the rate constants of ROO isomerization are dependent on three factors: ring strain energy, the nature of the broken C–H bond and the degeneracy of H atoms. Due to the presence of the ring structure, the H transfer involving cyclic carbons undergoes a bicyclic transition state, leading to higher strain energy compared with transition state of alkyl carbons. Therefore, such an estimation would lead to great uncertainty. In a recent study, Yao et al. [61] calculated the rate constants of isomerization for normal-alkyl cyclohexylperoxy radicals (up to n-butyl cyclohexylperoxy radical) at the CBS-QB3 level in combination with the transition state theory (TST) and Rice–Ramsberger–Kassel–Marcus/Master-Equation (RRKM/ME) theory. This kinetic dataset provided the high-pressure limit and pressure-dependent rate constants, and was used to prescribe the ROO isomerization in Plog format in the present model.

3.2.3. Reactions related to QOOH

As stated in the last section, the QOOH chemistry is crucial for the low temperature reactivity. In the well-established low temperature reaction scheme of alkanes, QOOH is mainly consumed through the second O_2 addition forming OOQOOH, decomposition to cyclic ethers and beta scission to alkenes. Here, the same reaction classes were applied to the construction of kinetic reaction mechanism for NBCH.

The reaction of oxygen molecule with QOOH (class 26) leads to the low temperature chain branching, and thereby is very important in the low temperature chemistry. The reaction constants for this reaction class were prescribed to be the same as the first O_2 addition, but the pre-exponent factors were divided by a factor of two according to the finds of Goldsmith et al. [62].

The decomposition of QOOH to the cyclic ether was believed to be one of the reaction classes responsible for the negative temperature coefficient (NTC) behavior [1,39,40,63]. Here, the formation of 3-,4- and 5-membered ring cyclic ethers was considered and the cyclic ethers were lumped based on the ring size. The rate constants were taken from multiple kinds of literature [46,57,64–67].

QOOH species with the radical site beta to hydroperoxyl group can undergo decomposition to produce a conjugated olefin and HO_2 radical. This reaction class is also responsible for the NTC behavior of hydrocarbons. Especially, the formation of conjugated olefins in the oxidation of cycloalkanes is a more important channel than in the oxidation of alkanes [52,53]. In this model, the conjugated olefins were lumped to two products, namely $C_4H_7cC_6H_{11}$ and $C_4H_9cC_6H_9$. For this reaction class, the rate constants for reactions involving the carbons numbered 5, 6, 7 and 8 were extrapolated from the reactions of the counterpart sites in the MCH model [41]; while for the other reactions, the rate constants were estimated based on those proposed by Bugler et al. [57]. QOOH species with the radical site gamma to hydroperoxyl group can undergo beta scission to form OH, carbonyl and alkene. The rate constants of this reaction class were assigned according to Bugler et al. [57].

It is the isomerization of OOQOOH to form ketohydroperoxide and an OH radical, and the subsequent decomposition of ketohydroperoxide to carbonyl radicals and OH radical (classes 27 and 28), that finalized the low temperature chain branching process. Here, the rate constants for isomerization of OOQOOH to ketohydroperoxide and OH were prescribed using those for isomerization of ROO to QOOH with some necessary modification according to the proposal of Sarathy et al. [39] and Bissoonauth et al. [41]. Specifically, the activation energies for the reactions not involving the ring structure were reduced by 3 kcal mol^{-1}, while for those involving the ring structure a reduction of 2 kcal mol^{-1} was applied. The A-factors were also adjusted to account for the number of available H atoms for transfer. The activation energy of decomposition of ketohydroperoxide in literature [39,48,49,68,69] ranged from 39 to 44 kcal/mol, which were close to the O–O bond strength of the ketohydroperoxide (44.5 kcal) [49]. Here, we adopted 42.5 kcal/mol, a value between the literature data.

3.2.4. Concerted elimination and other reaction classes

The concerted elimination reaction (class 16) eliminates an HO_2 radical from the ROO to yield conjugated olefins via a 5-membered transition state. It is also considered to be one of the reaction classes responsible for NTC behavior of alkanes. This type of reaction was included in the cyclohexane model proposed by Silke et al. [70] and found to be important for IDT predictions below 1000 K. Here, the conjugated olefins were lumped to two compounds, i.e., $C_4H_7cC_6H_{11}$ and $C_4H_9cC_6H_9$. For reactions yielding $C_4H_9cC_6H_9$, the rate constants were assigned to be the same as the counterpart of the MCH model [41]; while for reactions yielding $C_4H_7cC_6H_{11}$, we prescribed the rates as the analogous reactions of the isopentane model proposed by Bugler et al. [57].

For the two types of conjugated olefins, the sub-mechanisms were taken from the decalin model developed by Zeng et al. [71]. For reaction classes 12, 13, 14, 17, 18, 19, 20, 21 and 22 (not shown in Fig. 2), the reaction scheme was taken from the decalin model proposed by Wang et al. [47].

Generally, since the published studies on NBCH were rare, many rate constants in the present model were prescribed referring to analogous reactions for other cycloalkanes and some alkanes. Therefore, there is a necessity to further perform fundamental experimental and computational studies for the sake of developing a more accurate kinetic model. For the new species in the NBCH sub-mechanism, parts of the thermodynamic data were taken from literature [19,21,47], and others were derived using the group additivity method. It must be noted that further investigations are needed to provide higher accuracy model.

4. Results and discussion

4.1. Ignition delay times in the RCM

IDTs are an important global property of fuels influencing the performance of engines. Therefore, it is widely used as a validation target for a newly proposed kinetic model. Since the promising low temperature combustion concept relies on very lean or extremely diluted combustion [72], it is of great importance to assess the model against the IDTs of fuels under diluted conditions. Hong et al. [23] investigated the IDTs of NBCH in high temperature (1280–1480 K) under low pressures (1.5 and 3 atm). Here, we extend the work to the extremely diluted mixtures in low-to-intermediate temperature ranges at elevated pressures.

Figure 4 presents the measured IDTs of NBCH under different pressures. On the whole, typical NTC behavior is observed for NBCH under all testing conditions. In Fig. 4, the Arrhenius-type plots show that the reactivity increases with increasing pressure. The pressure dependence of IDTs in the NTC region become more pronounced than in low temperature range, due to the increasing importance of the pressure-dependent reactions in NTC region, e.g. H_2O_2+(M)=2OH+(M). Similar phenomena were observed in other fuels [29,32]. To evaluate the performance of kinetic models, both constant volume and variable volume simulations from the proposed model are also shown in Fig. 4. Generally, the proposed model follows the trends of the measured IDTs well under the testing conditions. It reliably captures the NTC behavior of NBCH, as well as the dependence of IDTs on pressure and equivalence

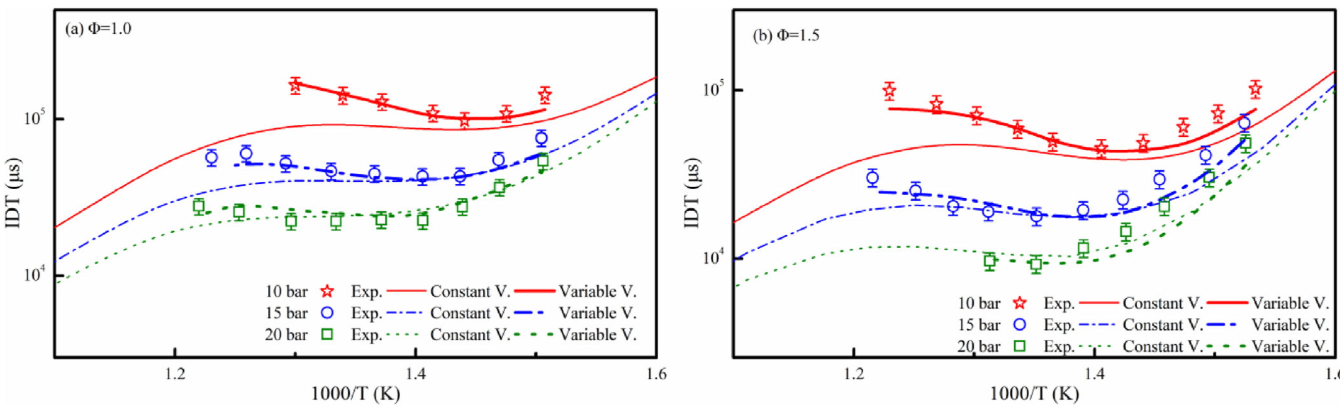


Fig. 4. Dependence of RCM-measured IDTs for NBCH on pressure for $\phi = 1$ and 1.5. Symbols: experiments; lines: simulation.

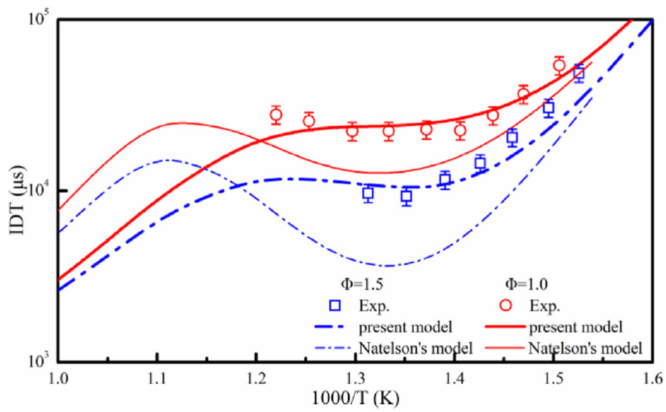


Fig. 5. Comparison of predictions from the present model and Natelson's model at 20 bar.

ratio. Overall, the predictions are in good agreement with the experimental results, basically within the error bar of 10%.

Figure 5 shows a comparison of predictions from the present model and Natelson's model [21]. Obviously, Natelson's model predicts much shorter IDTs, as well as a much pronounced NTC behavior than the experimental data. Overall, the present model provides better predictions.

4.2. Oxidation speciation in a flow reactor oxidation

In the present study, mole fraction profiles of oxygen and several key intermediate products were measured as a function of the initial reaction temperature (in 25 K steps) at 1 atm. The mole fraction of fuel was kept constant at 0.69% and the equivalence ratio was adjusted by varying the mole fraction of oxygen. In order to avoid thermal runaway, the test fuels were diluted with argon to approximate isothermal reaction conditions during oxidation. Figures 6–8 depict the measured mole fraction profiles of oxygen, as well as several major products. As a comparison, the predicted results from the present model and Natelson's model [21] are also shown.

Figure 6 presents the concentrations of oxygen, H_2, CO and CO_2 versus temperature. Out of them, oxygen acts as the oxidizer and thereby its consumption is considered as an index of system reactivity. We note that in Fig. 6(a) oxygen profiles at both equivalence ratios show a slight drop around 775 K, implying the very narrow low temperature reactivity window of NBCH at atmospheric pressure. Note that the low temperature reactivity in the present conditions (at 1 atm) is much weaker than in

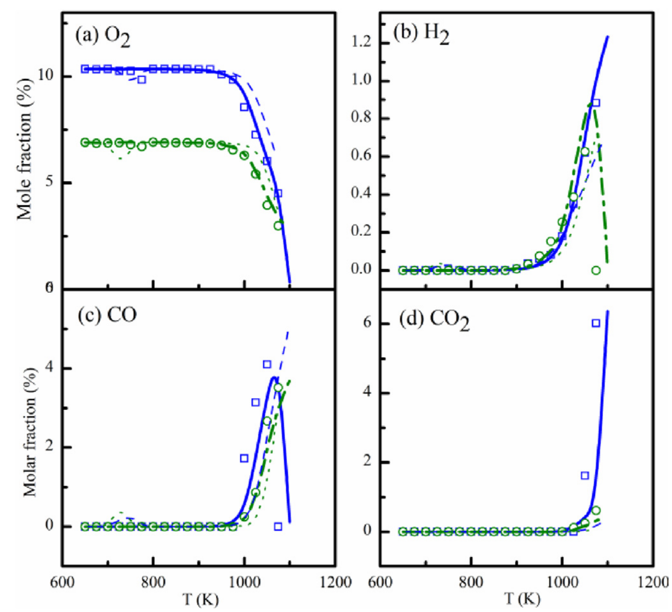


Fig. 6. Experimental (symbols) and simulated (lines) concentration of oxygen, hydrogen and main oxygenated products. Solid: predictions by the present model for $\Phi = 1.0$; Dash dot: predictions by the present model for $\Phi = 1.5$; Dash: predictions by Natelson's model for $\Phi = 1.0$; Dot: predictions by Natelson's model for $\Phi = 1.5$.

the experiments of Natelson et al. [21], which were carried out at 8 atm. As the temperature continues to increase, the mole fractions of oxygen increase and soon stay almost stable, implying a negative temperature coefficient behavior. At above 925 K, the mole fractions of oxygen begin to decrease rapidly, suggesting the starting of high temperature reactivity. It is found that the equivalence ratio has a significant effect on the reactivity, as the oxygen profile of the stoichiometric mixture shows a much steeper slope than the fuel-rich mixture. In general, the model satisfactorily captures the evolution of oxygen concentration and provides better predictions than Natelson's model [21]. The formation of H_2 occurs above 925 K and seems to be more favored for fuel-rich mixtures. CO is an effective indicator of low temperature chemical reactivity, since it is noticeably formed through the decomposition of aldehydes in the low temperature regime [51,73–75]. This is true according to the rate of production (ROP) analysis. In addition, the decomposition of ketohydroperoxide also plays an important role in the formation of CO. Consistent with the very weak low temperature reactivity suggested by the oxygen profiles, no detectable CO is formed at low temperatures, as shown in

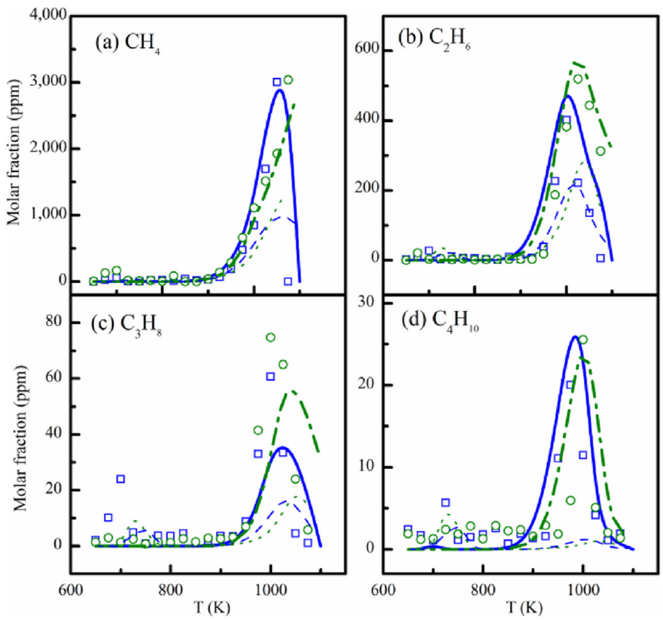


Fig. 7. Experimental (symbols) and simulated (lines) concentration of C1–C4 alkane products. Solid: predictions by the present model for $\Phi = 1.0$; Dash dot: predictions by the present model for $\Phi = 1.5$; Dash: predictions by Natelson's model for $\Phi = 1.0$; Dot: predictions by Natelson's model for $\Phi = 1.5$.

Fig. 6(c). In Fig. 6(d), the major final product CO_2 is mainly formed at high temperatures, while no detectable CO_2 was formed at low temperatures. The ROP analysis shows that CO_2 is mainly formed through the oxidation of CO. Combining Fig. 6(c) and (d), it can be found that CO oxidizes to CO_2 dramatically with the onset of high temperature chemistry. Hence, CO_2 peaks just after CO. Obviously, as shown in Fig. 6(d) that the formation of CO_2 is more favored in stoichiometric mixtures than in fuel-rich mixtures due to more oxygen available. Overall, the present model reasonably reproduces the evolution of the above-mentioned species, only slightly overestimates the formation of H_2 for fuel-rich mixtures.

Figure 7 presents the evolution of C1–C4 n-alkanes with temperature. Among them, methane is the most abundant alkane product, with a maximum of 3040 ppm at 1075 K for fuel rich mixtures, while ethane is also formed in considerable quantities. By contrast, the other two are in trace amounts, partly due to their poorer thermal stability. In general, compared with Natelson' model, the present model shows obvious better predictions. It reasonably predicts the evolution of all these alkanes with temperature, although it underpredicts the mole fraction of C_3H_8 by about 50%.

In this study, several alkenes with large amounts are detected. Figure 8 shows a comparison between the experimental and predicted results for three dominant alkenes, including C_2H_4, C_3H_6 and 1,3-C_4H_6. Obviously, C_2H_4 is the most dominant species of these alkenes. For the fuel-rich mixtures, it peaks at 1050 K with a maximum mole fraction of 7517 ppm, which is almost 10 times higher than the other two species. This roots in the better thermal stability and multiple formation pathway of C_2H_4, for instance, following the unimolecular decomposition of NBCH, the radicals C_4H_9, C_3H_7, C_2H_5 can easily decompose to produce C_2H_4 via beta scission or H elimination. Overall, the prediction from the present model is in good agreement with the experimental data.

4.3. Comparison of model prediction with literature data

This section evaluates the capacity of the newly constructed mechanism to predict different combustion characteristics of interest for NBCH oxidation by comparing the predictions with the published experimental data. Here, the experimental database used for evaluation emphasizes on oxidation environments and includes: (1) IDTs covering wide ranges of pressures, temperatures and equivalence ratios in ST and in RCM; (2) speciation profiles of oxidation in JSR.

4.3.1. Ignition delay times of NBCH

The available IDT measurements of NBCH include the ST studies by Hong et al. [23] and Zhang et al. [22], and our very recent study [27] using a ST and RCM. The three datasets were compared to the proposed model. In this section, both simulation of ST and RCM experiments were performed with constant volume constraint using Closed Homogenous Batch Reactor module of Chemkin-Pro software. The simulation results are presented in the form of IDTs of NBCH in air and under diluted conditions, respectively.

4.3.1.1. Ignition delay times of NBCH in air. Recently, Zhang et al. [22] measured the IDTs of NBCH in a ST spanning the temperature of 707–1458 K for equivalence ratio of 0.5, 1.0 and 2.0 under pressures of 2.0, 5.0 and 15 atm. Figure 9 shows a comparison of this dataset with the simulation results using the present model. In Fig. 9, error bars of ±20% as Zhang et al. [22] suggested was provided to evaluate the model. Generally, the model follows the trends of reactivity very well. Specifically, the model does well in capturing the dependence of IDTs on pressure and temperature. Especially, the NTC behavior of NBCH along with the slope of IDTs, as shown in Fig. 9(a) and (b) are well predicted by the model. Overall, the model reproduces the measured IDTs at a satisfactory level, although the model overpredicts the IDTs by 30–50% at several conditions.

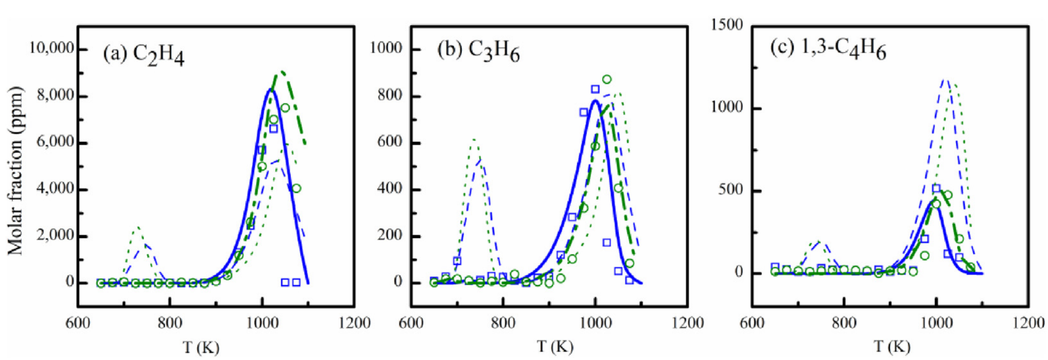


Fig. 8. Experimental (symbols) and simulated (lines) concentration of small alkene products. Solid: predictions by the present model for $\Phi = 1.0$; Dash dot: predictions by the present model for $\Phi = 1.5$; Dash: predictions by Natelson's model for $\Phi = 1.0$; Dot: predictions by Natelson's model for $\Phi = 1.5$.

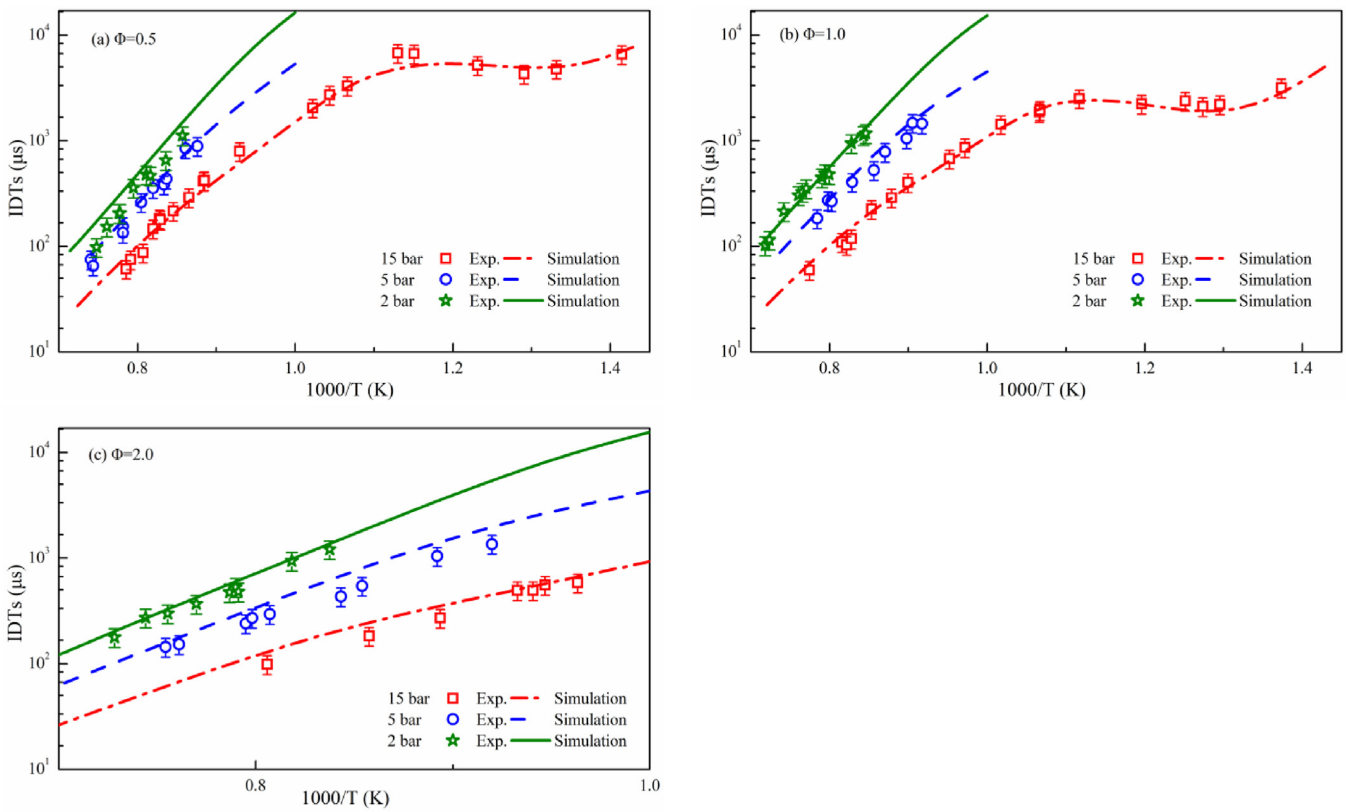


Fig. 9. Comparison of IDTs between predictions from the present model and the measured data [22].

In a recent study, we measured the IDTs of NBCH/air mixtures [27] under engine-like conditions. This dataset included equivalence ratios of 0.5, 1.0 and 1.5, pressures of 10, 15 and 20 bar, and temperatures of 612–1374 K. The proposed model was assessed against this dataset for the purpose of evaluating its performance at engine-relevant conditions. Figure 10 shows a comparison between the predictions and the experimental results. Overall, the model satisfactorily reproduces the IDTs of NBCH and the variation trends of IDTs revealed by the experimental results. For the tested mixtures, almost all the simulated IDTs fall within the experimental uncertainties of the measurements. Under all simulated conditions, the NTC behavior of NBCH was well captured by the model. In low and high temperature regions, the IDTs of NBCH show very weak dependence on pressure, with the strongest dependence on pressure in the NTC region. This trend is well captured by the model. However, discrepancies exist under some conditions. For fuel-lean mixture in Fig. 10(a), the model slightly overpredicts the IDTs in the NTC region at 10 bar. While for fuel-rich mixture in Fig. 10(c), the model slightly underpredicts the IDTs in the NTC region at 20 bar. These discrepancies demonstrate that further fundamental studies are needed to obtain more accurate kinetic rates.

4.3.1.2. Ignition delay times of NBCH under diluted conditions. The high temperature IDTs of NBCH at extremely diluted conditions have been measured by Hong et al. [23] in a ST. In their study, the mole fraction of O_2 was kept at 4.0% in an argon diluent for equivalence ratio of 0.5 and 1.0 at pressures of 1.5 and 3 atm. This dataset is used to evaluate the proposed model as well. Figure 11 presents the comparison between the predicted IDTs from the present model and experimental data. It is found that the model captures the trend of IDTs, as well as the slope of high temperature IDTs. Overall, the present model satisfactorily follows the autoignition characteristics of n-BCH, although it slightly overpredicts the IDTs for equivalence ratio of 0.45 at 1.5 atm.

Another IDT measurement under diluted conditions for NBCH was performed in our previous study over low-to-high temperature range at 10, 15 and 15 bar for stoichiometric mixtures with a dilution ratio of 89.5% ($\chi_{N2}/(\chi_{N2}+\chi_{O2})$) [27]. Figure 12 compares this dataset with the simulation of the model. The predictions show good agreement with the experimental data at all temperatures of interest.

In summary, the model can provide reasonable predictions for the IDTs of NBCH over wide conditions.

4.3.2. Comparison with speciation profiles of NBCH oxidation

Herbinet et al. [24] investigated the oxidation of NBCH in the JSR at Nancy under 1atm, at a residence time of 2 s, at temperatures ranging from 500–1100 K for three equivalence ratios of 0.25, 1 and 2. In this work, their experimental data was simulated using the Perfectly Stirred Reactor (PSR) module of the Chemkin-Pro software. The experimental and simulated mole fraction profiles of the main oxidation products are displayed in Fig. 13. In general, for most of the major species, the present model reproduces their evolution with the temperature at a satisfactory level. Both the reactivity of NBCH at low temperature and high temperature region are well captured by the model. Specifically, as shown in Fig. 13(a) and (b), the mole fraction profiles of oxygen and NBCH exhibit a typical low temperature reactivity and NTC behavior, which is well predicted by the present model. In addition, this model reliably mimics the dependence of reactant consumption on equivalence ratio. It is well established that CO is a good indicator of low temperature reactivity. The present model satisfactorily captures the variation of CO of low temperatures in Fig. 13(c), demonstrating the reasonability of the low temperature reaction scheme. The model also accurately predicts the trends of final oxidation product CO_2, as shown in Fig. 13(d). The comparisons of the simulated C1–C2 alkane profiles with the experiments are shown in Fig. 13(e) and (f). Generally, the model follows the experimental

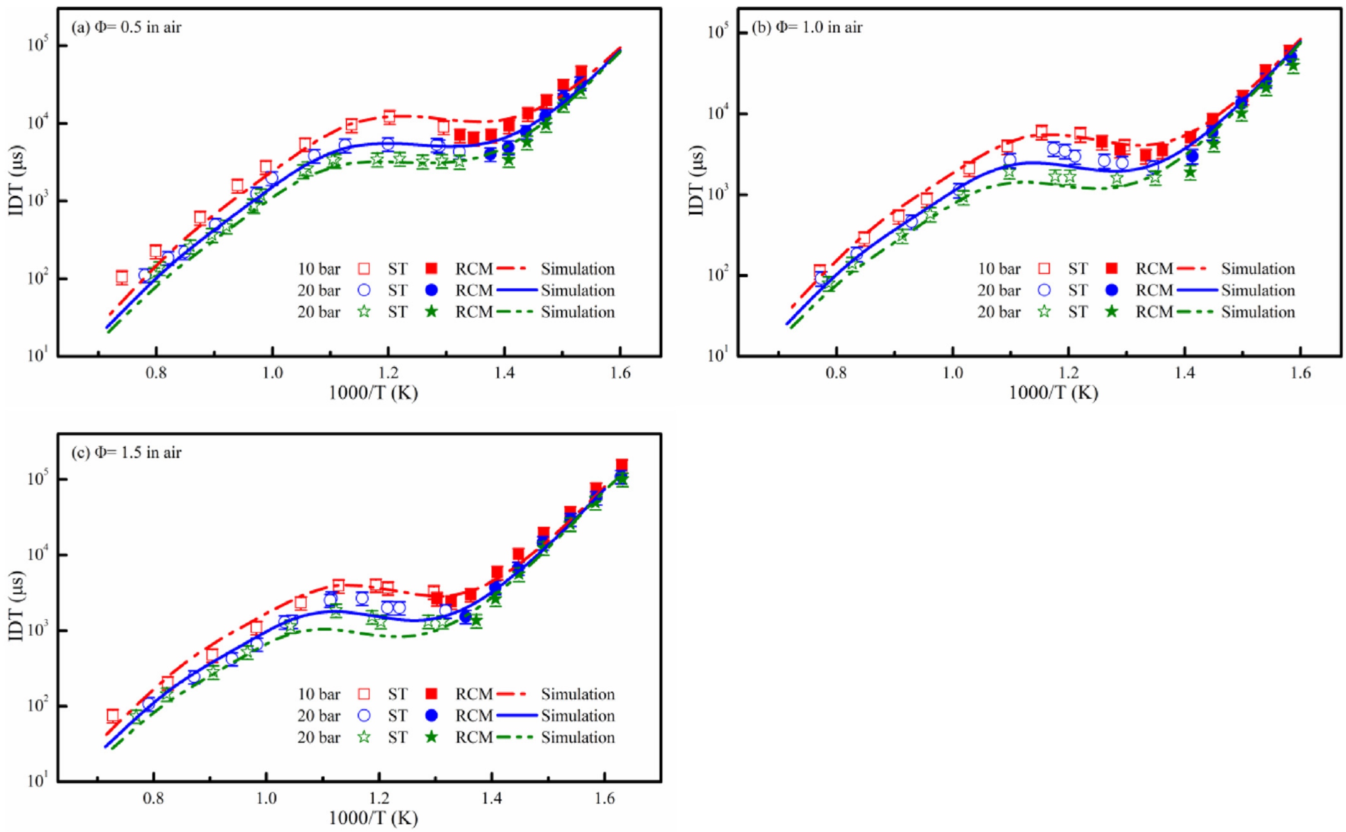


Fig. 10. Comparison between experimental and simulated IDTs of NBCH in the air at various temperatures, pressures, equivalence ratios [27].

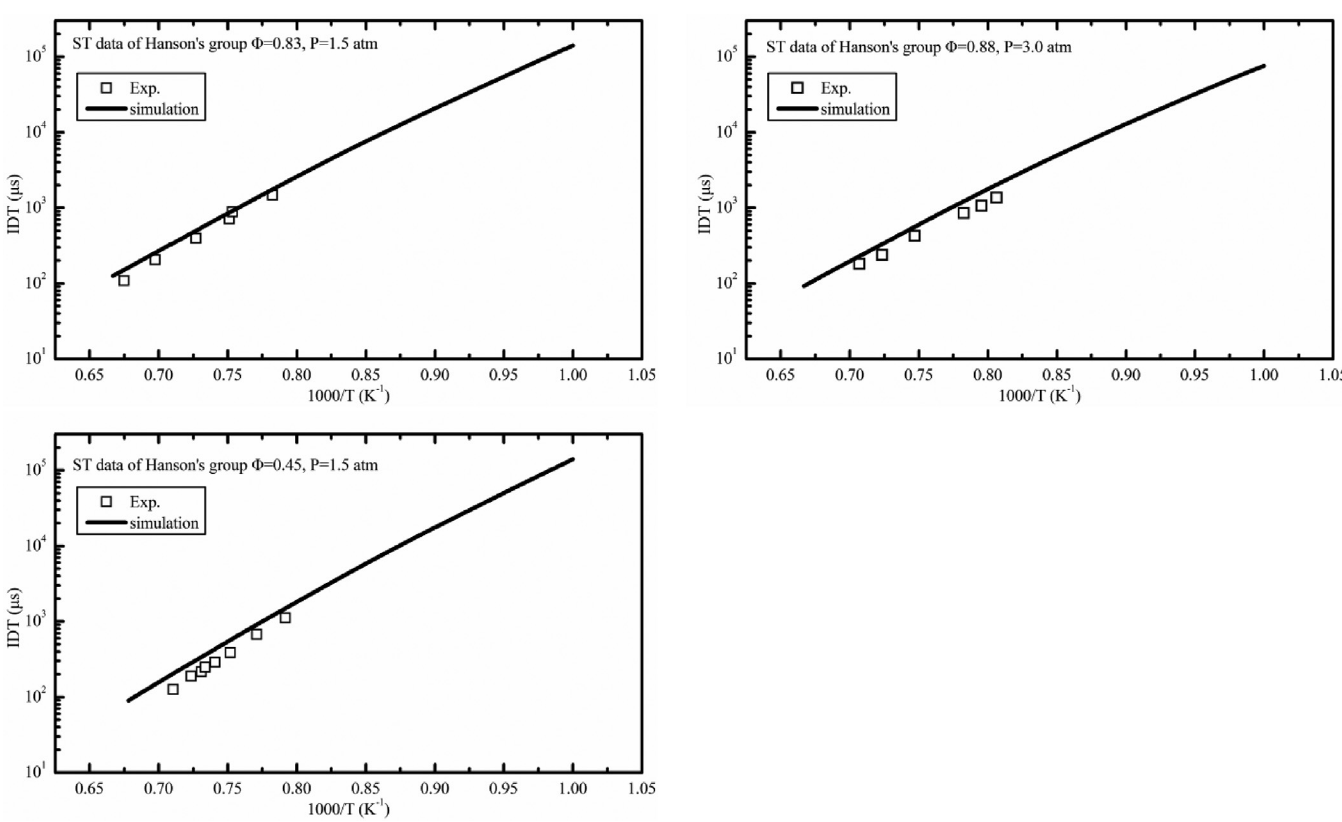


Fig. 11. Simulated and experimental IDTs of n-BCH in a ST for diluted mixtures [23].

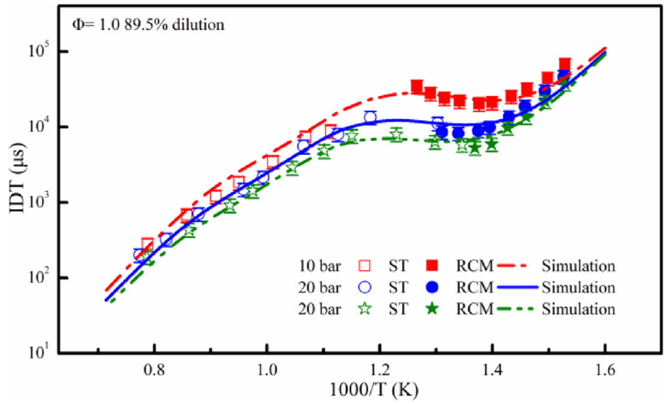


Fig. 12. Simulated and experimental IDTs of n-BCH in a RCM and ST for diluted mixtures [27].

behavior very well over the entire temperature region. C_2H_4 was selected to represent the alkenes and simulated, as shown in Fig. 13(g). The model provides excellent prediction at high temperature region, however, it overpredicts the concentration of C_2H_4 at low temperature region. Figure 13(h) and (i) illustrates the comparison between the simulated and experimental data of several species closely related to pollutant emission, including acetaldehyde and benzene. The model well predicts the formation of acetaldehyde at low temperature, but underestimates its formation at high temperature range. The model reproduces the speciation profiles of benzene at an acceptable level for fuel-lean and stoichiometric mixtures, but fails to do so for fuel rich mixture. Generally, the agreements are reasonably good under most operating conditions with consideration of the experimental uncertainties.

4.4. Reaction flux analyses and sensitivity analyses

In order to elucidate the reaction pathways that drive NBCH reactivity, reaction flux analysis was performed for stoichiometric NBCH/air mixtures at low (650 K), intermediate (800 K) and high (1300 K) initial temperatures at an initial pressure of 20 bar. The analyses were conducted at the time corresponding to 20% fuel conversion, because it is representative in the process of fuel oxidation and the reactions tend not to be affected significantly by thermal runaway.

Figure 14 shows the results of the flux analysis. It is observed that from low to high temperature, the parent fuel is mainly consumed through H atom abstraction by active radicals, among which OH, H and HO_2 contribute the most. At 650 and 800 K, OH radical contributes most (96.8% and 87.7%, respectively) to

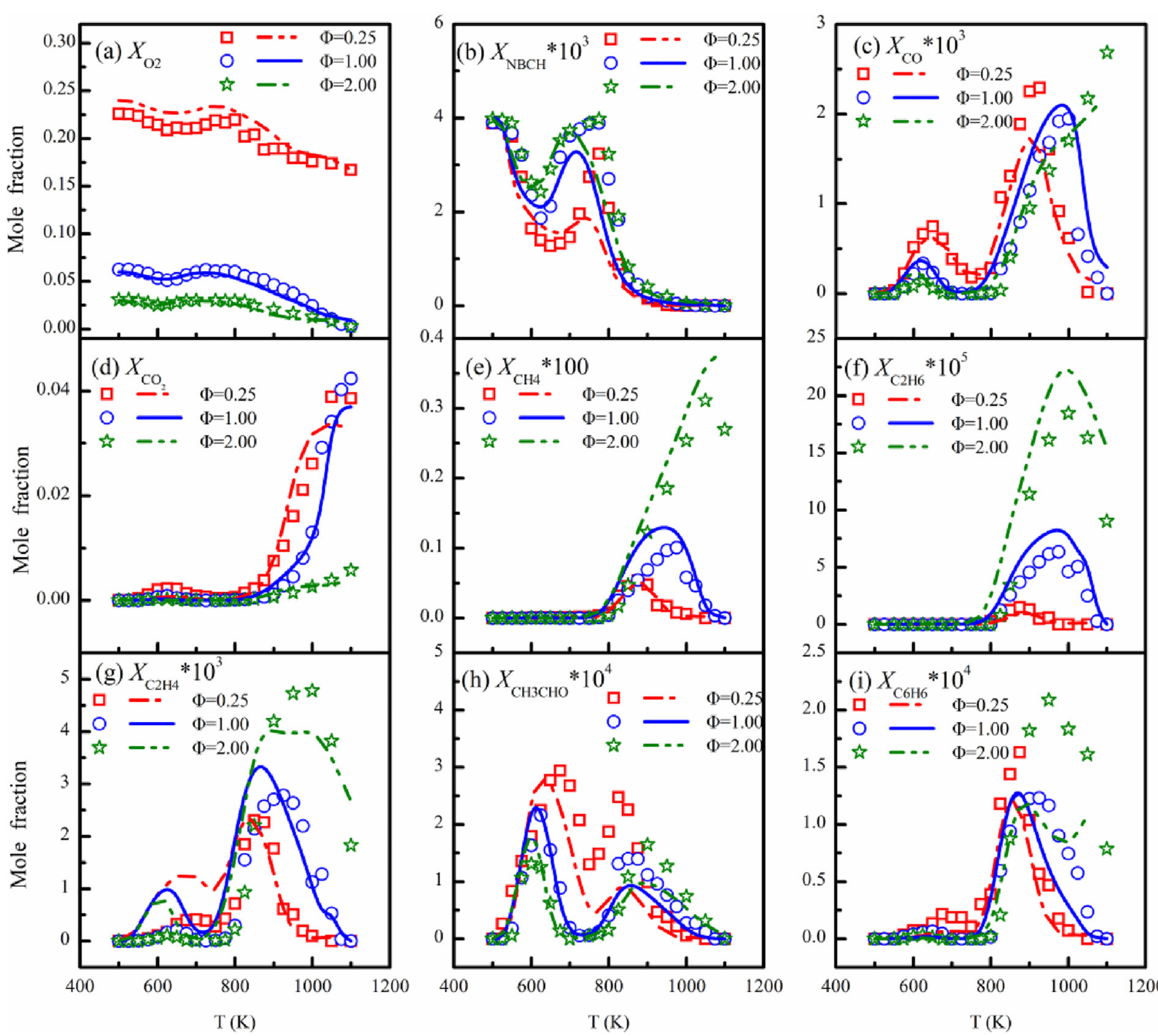


Fig. 13. Experimental and simulated mole fraction profiles of NBCH, O_2 and major oxidation species in the JSR [24].

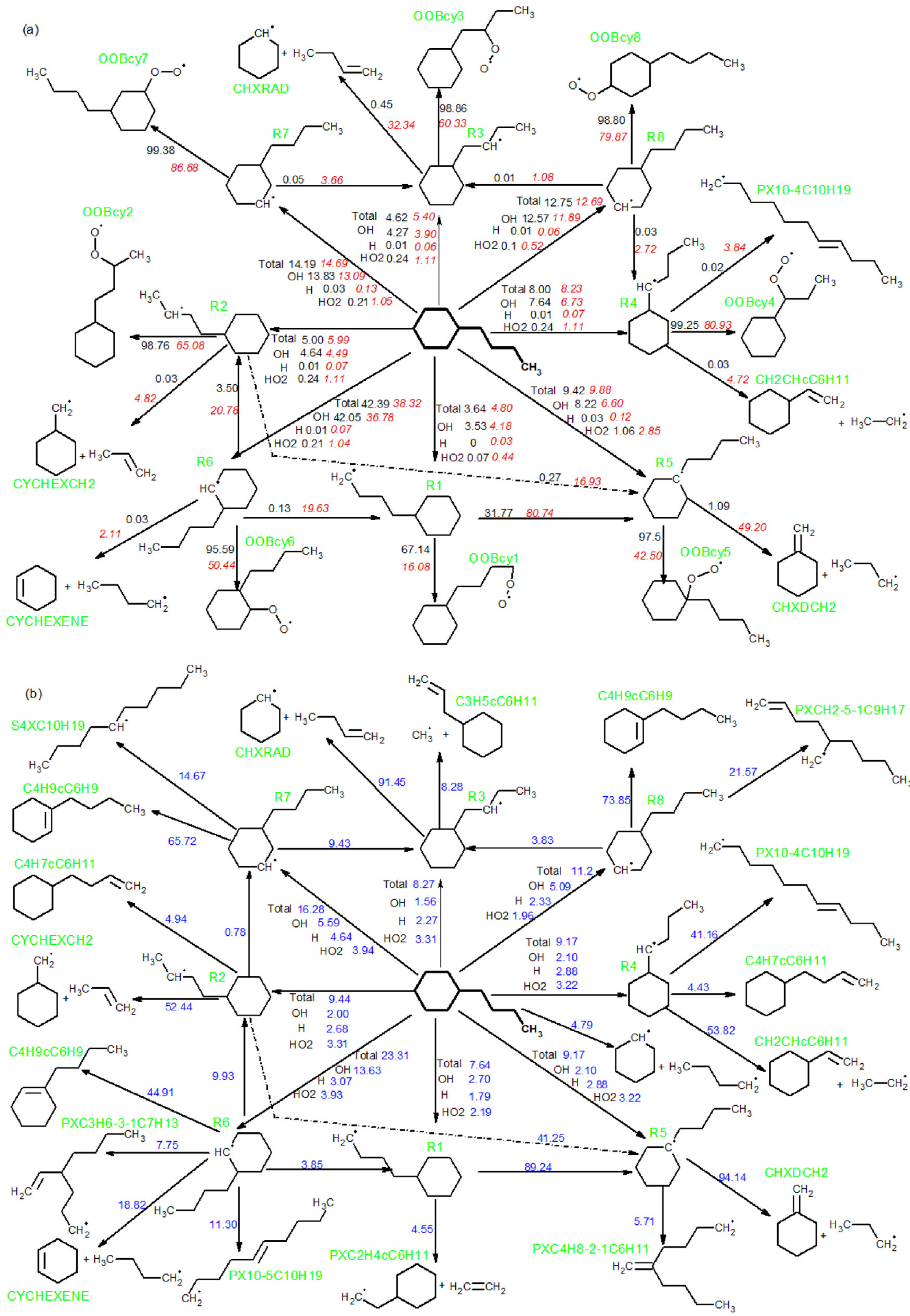


Fig. 14. Flux analysis for NBCH oxidation under 20 bar for the stoichiometric mixture at the time of 20% fuel conversion. (a) Initial temperatures of 650 K (normal) and 800 K (italics) and (b) initial temperatures of 1300 K (blue). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

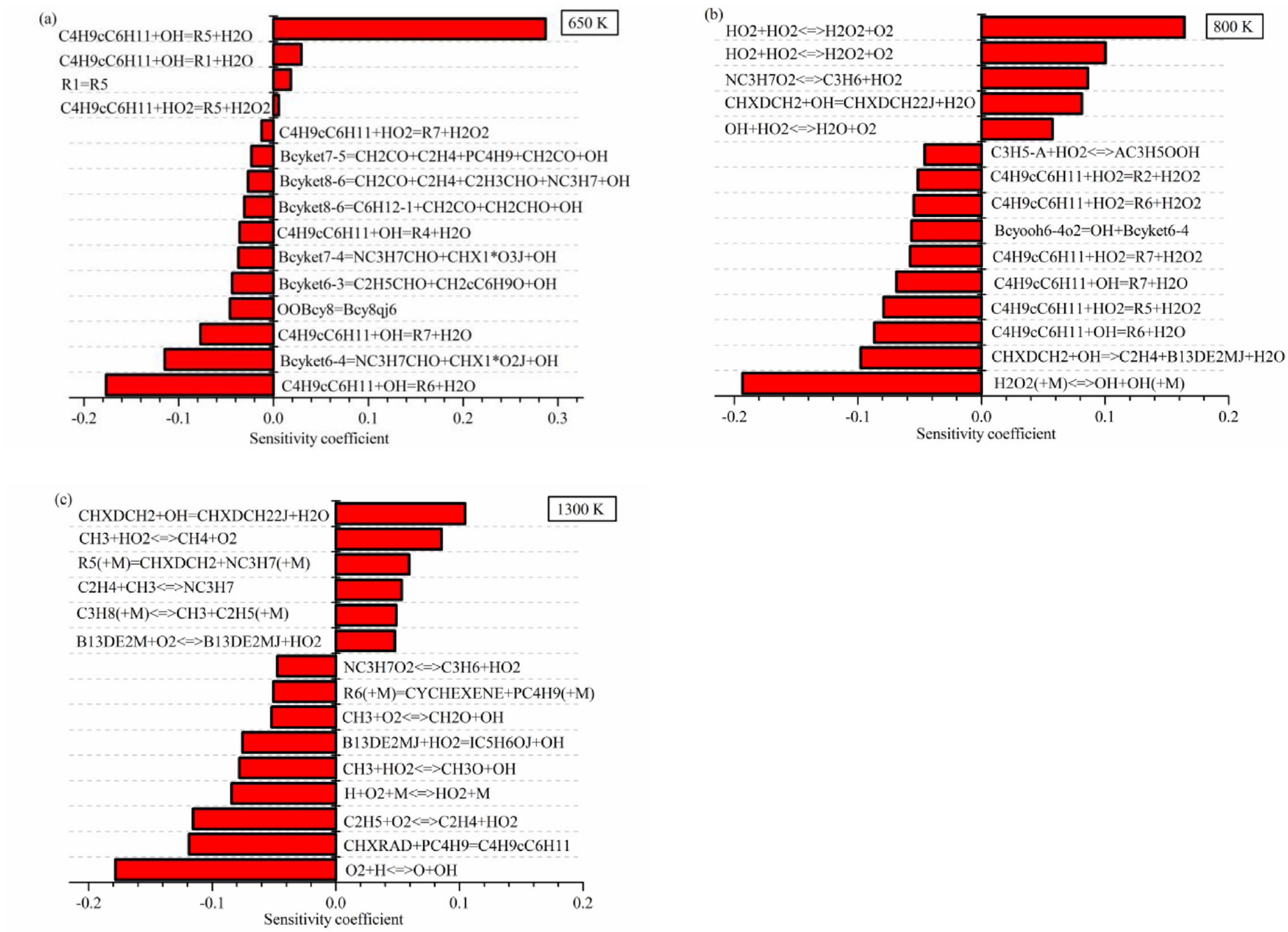


Fig. 15. Brute force sensitivity analyses performed at a pressure of 20 bar for the stoichiometric mixture at initial temperatures of 650, 800 and 1300 K.

the fuel consumption. However, the significance of H and HO_2 increases as temperature increases, and they consume comparable fuel with OH radical at 1300 K. Therefore, at high temperatures the H-abstraction by H radical begins to compete with the chain branching reaction $H+O_2$=O+OH more effectively, leading to a crossover between the reactivity of mixtures with different equivalence ratios [27,76]. At 650 and 800 K, almost all fuel is consumed by H abstraction reactions, while at 1300 K, the fuel consumed by unimolecular decomposition increases markedly, contributing 4.79% of fuel conversion. Note that unimolecular decomposition mainly happens via C–C scission in the substitute chain, while the ring-opening reactions which involve the cyclic C–C bonds are negligible. By contrast, H-elimination channels are negligible at entire temperature range due to the high activation energy and the low entropy of the variational transition states [77,78].

The consumption channels of the fuel derived radicals are quite different in low temperature and high temperature ranges. The O_2 addition, which initiates the low temperature reaction sequence, is not very important for R1 conversion at 650 K, with only a contribution of 67.14%. However, this reaction type is crucial for the consumption of the other radicals at low temperatures, specifically, more than 95% of these radicals involved O_2 addition reactions. Compared with O_2 addition, the beta scission channels are negligible in the low temperature range. As temperature increases, the contribution of O_2 addition decreases and the beta scission channels play an increasingly important role. At 800 K, the beta scission channels consume a considerable portion of the radicals. For instance, 32.34% of R3 and 49.20% R5 are conversed by beta scission channels. At 1300 K, except the isomerization reactions between each other via intra-molecule H transfer, the fuel derived radicals mainly undergo beta-scission. For R1–R5, the radicals mainly undergo C–C bond scission to produce an olefin and a small radical or to produce an alkenyl through ring-opening at 1300 K. While for R6–R8, the channel of C–H bond scission to produce cyclic alkene (lumped as $C_4H_9cC_6H_9$, which is considered to be precursors of aromatics,) is more favored. By comparing Fig. 14(a) and (b), it can be found that the consumption channels of fuel to produce radicals R6–R8 are preferred, implying that NBCH would tend to show strong trend towards soot formation, just as other cycloalkanes in literature [15,50,52,79,80].

In order to identify the key reactions that dominate the oxidation chemistry of NBCH, a brute force sensitivity analysis was performed for equivalence ratio of 1.0 at 20 bar. The sensitivity was defined as: $S_i = [\tau(2k_i) - \tau(k_i)]/\tau(k_i)$. Here, k_i represents the rate constant of reaction *i*, $\tau(k_i)$ and $\tau(2k_i)$ denote the predictions when the rate constant of reaction *i* was set as k_i and $2k_{i,}$ respectively. Three temperatures, 650, 800 and 1300 K, were selected to represent low, intermediate and high temperature ranges, respectively.

Figure 15 shows the top 15 sensitive reactions at the three temperatures. At 650 K, the H abstraction from NBCH and the decomposition of ketohydroperoxide are the most important reactivity promoting reactions. In addition, isomerization of ROO also plays an important role in promoting reactivity. We note that the H abstraction from NBCH to form R5 shows the most inhibiting effect. This is because after the second O_2 addition, the OOQOOH (derived from R5) could not isomerize to form ketohydroperoxide because there is no H atom to be abstracted on a tertiary carbon. In addition, among all the fuel derived radicals, R5 has the most

H atoms on the beta carbons, hence, the concerted elimination of OOBcy5 is most favored. Consequently, the low temperature degenerate chain branching sequence of this channel would be interrupted and thereby the system reactivity would be suppressed. Note that in Fig. 14, a considerable part of R1 undergoes isomerization to produce R5. What's more, the strong primary C–H makes this channel less preferred. Therefore, the reactions producing R1 show inhibiting effect as well. At 800 K, the reactions of H_2O_2 control the system reactivity. The decomposition of H_2O_2 shows the highest promoting effect, while its competing reactions $HO_2+HO_2=H_2O_2+O_2$ (duplicate reactions) show the highest inhibiting effect. H abstraction from NBCH still shows high sensitivity to the system reactivity. However, the abstractor HO_2 plays a more important role than OH radical due to the increasing significance of H_2O_2 in this temperature range. This is consistent with the increasing importance of HO_2 in consuming NBCH in Fig. 14 with increasing temperature. At 1300 K, the high temperature chain branching reaction O_2+H =OH+O dominate the overall reactivity. Note that the unimolecular decomposition strongly promotes the reactivity, which is consistent with the increasing importance of this reaction class in high temperature range shown in Fig. 14.

5. Conclusions

A detailed kinetic model was developed to describe the combustion chemistry of NBCH from low-to-high temperature range in this work. The autoignition and oxidation characteristics of NBCH were investigated in a RCM and flow reactor over a wide range of conditions, respectively. The IDTs of NBCH under extremely diluted conditions were measured at low-to-intermediate temperature range at 10, 15 and 20 bar. Numerous oxidation species were detected and quantified in the flow reactor oxidation. The present experimental data was used as a target for the proposed model. Good agreement between the experimental and simulated data was observed. The model was further compared to available experimental datasets of NBCH in recent literature, including IDTs measured in RCM and ST, oxidation species profiles in JSR. The good performance of the model adds substantially to our understanding of the NBCH chemistry, and it is anticipated to be a starting point for the development of chemical kinetic reaction mechanisms for longer substituted alkylcyclohexane. Since the kinetics of NBCH was scarcely studied previously, the rate coefficients of several key reactions were assigned based on rate rule or analogies, and thus potentially have great uncertainties. Therefore, further improvements can be done to this model with the availability of more accurate rate constants and experimental datasets in future work.

Acknowledgment

The authors gratefully acknowledge the financial support from the Key Project of National Natural Science Foundation of China (Nos. 91641202 and 51425602) and Program of Shanghai Subject Chief Scientist (No. 19XD1401800).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.09.007.

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